

Uncertainty Modelling in Under-Represented Languages with Bayesian Deep Gaussian Processes

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Abstract

NLP models often face challenges with under-represented languages due to a lack of sufficient training data and language complexities. This can result in inaccurate predictions and a failure to capture the inherent uncertainties within these languages. This paper introduces a new method for modelling uncertainty in under-represented languages by employing deep Bayesian Gaussian Processes. We develop a novel framework that integrates prior knowledge and leverages kernel functions. This helps enable the quantification of uncertainty in predictions to overcome the data limitations in under-represented languages. The efficacy of our approach is validated through various experiments, and the results are benchmarked against existing methods to highlight the enhancements in prediction accuracy and measurement of uncertainty.

1 Introduction

Under-represented languages (Midrigan-Ciochina et al., 2020), spoken by geographically marginalized communities facing significant environmental or social challenges, present a unique opportunity for NLP (Ignat et al., 2024). By analyzing local news and social media data in these languages, NLP models can aid in disaster response, resource management, and public health initiatives (Lankford et al., 2023). Besides, NLP models can empower these communities by facilitating communication (Rodríguez et al., 2024), education, and economic opportunities in the digital age (Cissé and Sadat, 2024).

NLP has the potential to revolutionize communication and information access for everyone. The development of pre-trained large language models (Wang et al., 2022; Chang et al., 2024) has further amplified this potential. Including under-represented languages in NLP research allows us to explore the broader spectrum of human language

and communication, enriching our understanding of language itself. However, this promise has not been fully realized (Holgado and Vergez-Couret, 2024). A significant digital divide exists (Carthell, 2024), as many NLP models struggle in under-represented languages (e.g., Urdu and Pashto (Ali et al., 2024)) due to limited training data. This data scarcity hinders the models' ability to capture the nuances and complexities of these languages (Qin et al., 2024).

The limited data available for under-represented languages poses a two-fold challenge for NLP models. First, it hinders their ability to learn accurate language representations, leading to frequent prediction errors (Rambachan, 2024). Second, it makes it difficult to quantify the inherent uncertainty associated with these predictions (Kim et al., 2024). Uncertainty quantification (Combs et al., 2024), however, is crucial for understanding the reliability of model outputs and identifying areas where the model is less confident. For instance, consider addressing cybercrimes in Pakistan, where Urdu is the dominant language. With millions of social media users, manual monitoring is not feasible. Therefore, reliable automated methods are essential. Table 1 showcases example predictions from our model. We observe that the model's predicted probabilities tend to be high when it is confident in its classification. This suggests a strong correlation between confidence and high probability values. However, the key challenge arises when the model exhibits lower confidence (as reflected by lower predicted probabilities). By capturing this inherent uncertainty, we can identify areas for improvement and develop remedial measures to enhance the model's overall reliability.

Languages such as Urdu exhibit rich morphology, meaning words can have complex structures formed by adding prefixes and suffixes (Maaz et al., 2024). Traditional deep learning architectures might struggle to capture these intricate morpho-

Text [Translation]	GL	PL	PP	RP
Multi-class				
Meri shadi ho rai hai [I am getting married]	happy	happy	0.961	0.037
MQM K Rukn e Assembly Landhi Mn Awami Ahtijaj Mn Shamil 2 Maah Se Pani Nahi Dia Ja Raha Ahtijaj Na Krian Tu Kia Krian [MQM Member of the local assembly, have joined the public protest in Landhi, water has not been given for 2 months, if not protest then what else they can do]	sad	angry	0.402	0.348
Kyun ke wo acting ke sath fun e raqs mein bhi apna jawab nahi rakhti [Because along with acting, she has no match in dancing]	happy	neutral	0.525	0.454
Binary-class				
چیراں ہو جائے اگر میری ویڈیوز دیکھو گے تو [You will be surprised if you watch my videos]	neutral	neutral	0.876	0.124
چوتیا پا کرنے سے بہتر ہے پہلے جیسا کام کر تو کمزری سی کرتا ہے ادکاری نہیں [It is better for you to do the same fu**ing thing as before, you are good in commentary, not in acting]	offensive	neutral	0.515	0.485
Binary-class				
ہش کرو ، پیش تپا ہت گنوم یزخ بناؤز یدد ایب وا اییس یوش لژور باجنپ د پش رادرم مہ ہب ٹیخ اد [This bad guy will also die, the trained dog of Punjab, and then his youth will be left as a peanut, disappear]	offensive	offensive	0.906	0.094
یوروخ تخس یم لوک طایتحا ہن یوودد امر [It hurts me to be careful of them]	neutral	offensive	0.511	0.489

Table 1: Sample texts of the datasets with their ground-truth labels (GL), predicted labels (PL), predicted probabilities (PP), and runner-up probabilities (RP) by the BERT-multilingual model.

logical relationships (Anam et al., 2024). They also use complex writing systems with a character-based script (Aars et al., 2024). One obvious approach is to use transfer learning (Muraoka et al., 2023) using language models. The issue is that language models might encode biases present (Fang et al., 2024) in the dominant language data they were trained on. These biases can be transferred to the under-represented language model, leading to unfair or inaccurate results (Schwöbel et al., 2023).

Fine-tuning a language model for an under-represented language requires a careful selection of layers and hyperparameters to avoid overfitting (Dhananjaya et al., 2024) to the limited training data (Hangya et al., 2022). In this situation, Bayesian approaches such as Gaussian Processes (Marrel and Iooss, 2024) offer a promising approach to address some of the challenges associated with modelling under-represented languages such as Urdu and Pashto. Bayesian models can learn effectively from limited data (Xu et al., 2024) compared to deep learning models. This is because they focus on the relationships between data points rather than requiring a massive dataset to learn complex feature representations. These models allow us to incorporate prior linguistic knowledge about the target language as a prior distribution. This knowledge can be in the form of grammatical rules, word embeddings from other languages, or existing dictionaries. They offer a more flexible framework compared to traditional deep learning models. The choice of the kernel function (Buch, 2011) allows for customization to model different types of relationships between data points in the under-represented language. These models can be more interpretable than deep learning models.

Estimating uncertainty is vital for understanding and gauging the reliability of machine learning models. Typically, these models provide a single

prediction or point estimate (Gal and Ghahramani, 2016) without indicating confidence level. Uncertainty estimation offers a method to quantify such confidence. A high uncertainty value from a model suggests that its prediction may be less reliable. This is particularly important in scenarios where incorrect decisions could have serious repercussions. Moreover, uncertainty estimation helps evaluate model performance in various situations, such as when dealing with new or slightly varied data.

The **contributions** of the paper are: **1)** We introduce a novel model that extends deep Gaussian Processes by incorporating prior knowledge into a standard GP framework, enabling the model to capture complex relationships in under-represented languages and quantify uncertainty in predictions. **2)** A kernel function is employed to better capture and model intricate linguistic patterns and correlations, enhancing both the performance and confidence of predictions on complex datasets. **3)** The study also evaluates the effectiveness of various multilingual language models, specifically focusing on their performance with lesser-studied languages such as Urdu and Pashto, providing valuable insights for low-resource language processing. To the best of our knowledge, limited research has been done in modelling uncertainty in Urdu and Pashto.

2 Literature Review

Under-Represented Languages: Natural language processing has been transformed with the advent of pre-trained language models (Wang et al., 2022). Yet, a significant disparity persists between well-resourced and under-represented languages (Majewska et al., 2022). Under-represented languages are those with limited resources for developing digital language tools. They typically have fewer publicly available datasets and are underrep-

resented in pre-trained models, making it challenging to apply advanced NLP techniques compared to more widely studied languages. Efforts are underway to bridge this gap, [Lankford et al. \(2023\)](#) has adapted multilingual language models for low-resource languages. Similarly, various other approaches have been explored. For instance, [Winata et al. \(2022\)](#) used few-shot learning to perform sentiment analysis on 12 languages, including 8 languages not previously studied, to evaluate the effectiveness of various few-shot learning techniques. To enhance the performance of large language models for under-represented languages, [Ullah et al. \(2023\)](#) explored prompt engineering. Their findings show a significant improvement, with prompt-based fine-tuning leading to a 13% increase in accuracy over traditional fine-tuning methods. Addressing the issue of scarce data in under-represented languages, researchers have examined various data augmentation techniques. [Azam et al. \(2022\)](#) focused on strategies for Urdu, and [Khalid et al. \(2023\)](#) investigated these for the Punjabi language. While these methods are beneficial, the uncertainty in their outcomes remains a concern, particularly for complex languages such as Urdu and Pashto.

Deep Gaussian Processes: Deep learning and deep Gaussian Processes (dGPs) ([Williams and Rasmussen, 2006](#); [Griffiths et al., 2024](#); [de Souza et al., 2024](#)) are both influential machine learning techniques ([Lee et al., 2017](#); [Damianou and Lawrence, 2013](#)), sharing some similarities yet also having notable differences. Both methods use a layered architecture to extract increasingly complex features, enabling them to effectively handle intricate relationships and patterns. Both approaches can capture non-linear relationships between inputs and outputs, crucial for tackling complex real-world problems. Finally, both methods learn from data by adjusting internal parameters, allowing them to adapt to various tasks and datasets. However, for interpretability and uncertainty quantification, dGPs are often the preferred choice. The authors in [Dimitrakopoulos et al. \(2023\)](#) propose a technique using implicit neural representations for efficient Bayesian inference in low-dimensional problems. These methods are powerful tools for representing complex functions by learning to map input data into a high-dimensional latent space.

Uncertainty Modelling: The study by [Gal and Ghahramani \(2016\)](#) addressed uncertainty modelling by using dropout as a Bayesian approximation to represent uncertainty in deep learning mod-

els ([Yao et al., 2024](#)). This technique utilizes the randomness of dropout during training to gauge the range of possible predictions, thereby providing an uncertainty estimate for each output. [Tanner et al. \(2023\)](#) proposed two innovative metrics, Verbalized Uncertainty and Probing Uncertainty, to gauge the uncertainty of explanations generated by large language models. In ([Xiao et al., 2022](#)), the authors provide guidelines for developing a prediction pipeline based on pre-trained language models (PLMs) aimed at reducing calibration error. These recommendations are grounded in their empirical analysis of uncertainty quantification in PLMs. In ([Nehme et al., 2024](#)), the authors explore uncertainty quantification using neural posterior principal components.

[Watson et al. \(2024\)](#) contend that deep learning models typically achieve high predictive accuracy but lack transparency in their results. The opaque nature of deep learning models creates challenges in trusting and deploying them, especially in critical real-world applications such as detecting cybercrimes ([Ullah et al., 2024](#)). These black box models lack a way to quantify their output confidence. To address this, researchers have proposed various extensions and improvements building on Gal’s pioneering Monte Carlo dropout ([Gal and Ghahramani, 2016](#)). Nevertheless, a thorough literature review highlights ongoing gaps that our research seeks to fill. In this paper, we address these gaps by using priors alongside kernel functions in the deep Gaussian Processes model. Current Bayesian Monte Carlo methods frequently struggle to utilize prior information regarding the model’s parameters. Our model fills this void by introducing a systematic framework for integrating informative priors, resulting in more resilient and dependable uncertainty assessment.

3 Model Description

We outline the technical specifics of the deep Gaussian Process (dGP) ([Damianou and Lawrence, 2013](#)) model and then discuss the specifics of placing priors on its hyperparameters. Introducing priors to the dGP model’s hyperparameters enhances performance and interpretability. Key benefits include regularization, which penalizes significant deviations from the prior distribution. Without priors, hyperparameters begin at arbitrary values, often resulting in slower training and less optimal outcomes. Additionally, well-chosen priors improve

uncertainty estimates by reflecting our understanding or hypotheses about data variability, thus aiding in more accurate confidence calibration and dependable uncertainty quantification.

3.1 Deep Gaussian Processes Model

A dGP is an extension of a standard Gaussian Process (GP). It consists of multiple GP layers stacked together, each possibly employing a distinct kernel function, to model intricate data relationships. A dGP with L layers is represented as $f_L(x)$, where x signifies the input data and L indicates the layer index. Each layer applies a transformation to the output from the preceding layer or the input data in the case of the first layer utilizing its specific kernel function and corresponding hyperparameters.

The kernel function, represented as $k_l(x, x')$, is pivotal in each DGP layer l . It establishes the covariance between the outputs at two distinct input points, x and x' . Typical choices for kernel functions are RBF (Radial Basis Function) kernels or spectral mixture kernels. Each kernel function possesses a unique set of hyperparameters, symbolized as θ_l , which govern its characteristics. For example, an RBF kernel may include a lengthscale hyperparameter that dictates the rate at which the covariance diminishes as the distance between input points increases.

The output of a dGP layer l , $f_l(x)$, can be expressed recursively as:

$$f_l(x) = \mu_l(x) + K_l(x, X)[f_{(l-1)}(X) - \mu_{(l-1)}(X)] + \epsilon_l(x) \quad (1)$$

where, $\mu_l(x)$ denotes the mean function of layer l (commonly set to zero for simplicity), $K_l(x, X)$ represents the kernel matrix for layer l , with elements $k_l(x, x_i)$ corresponding to all training data points $X = \{x_1, \dots, x_N\}$, $f_{(l-1)}(X)$ is the vector of outputs from the preceding layer (or the input data for $l = 1$), $\mu_{(l-1)}(X)$ is the vector of mean function values from the preceding layer (often zero), and $\epsilon_l(x)$ is a white noise term that signifies noise in layer l .

3.2 Our Fully Bayesian dGP Model

A dGP is a probabilistic machine learning model that enhances the functionality of standard GP by incorporating multiple GP layers. Standard GPs often face challenges with high-dimensional data or complex feature interactions, such as those found in under-represented languages. dGPs overcome

Input: L as the number of dGP layers,
 $\mathcal{K} = \{k_1, k_2, \dots, k_L\}$ a list of kernel functions for each layer, $\Theta = \{\theta_1, \theta_2, \dots, \theta_L\}$ the sets of hyperparameters for each kernel,
 $\mathcal{P} = \{p(\theta_1), p(\theta_2), \dots, p(\theta_L)\}$ the prior distributions for hyperparameters, X the training data, y the training labels (for classification tasks).

Output: A trained dGP model

1. Define dGP Model:

- Initialize an empty dGP model.

for $l = 1$ **to** L **do**

 - Insert layer l into the model with kernel function k_l .

end

2. Hyperparameters and Hyper Priors:

for $l = 1$ **to** L **do**

 - Determine hyperparameters θ_l for kernel k_l .

 - Associate each θ_l in the model with its corresponding prior $p(\theta_l)$ from $\mathcal{P}$. If we place the Normal Inverse Gamma prior,

$$f_{NIG}(\theta_l | \alpha, \beta, \mu, \delta) = \frac{\alpha \delta \exp(\delta \sqrt{\alpha^2 - \beta^2} - \beta(\theta_l - \mu))}{\pi \sqrt{\theta_l} K_1 \left(\frac{\alpha \sqrt{\delta^2 + (\theta_l - \mu)^2}}{\delta \sqrt{\alpha^2 - \beta^2}} \right)}, \text{ where } \theta_l \text{ is the}$$

hyperparameter value, α, β, μ , and δ are the NIG distribution parameters, K_1 is the modified Bessel function of the third kind and order 1. While the NIG density function provides the probability density for a single value θ_l , the prior itself represents the entire distribution of possible hyperparameter values. This distribution is obtained by integrating the NIG density function over the entire valid range of the hyperparameter (often positive values for variance or scale):

$$p(\theta_l) = \int f_{NIG}(\theta_l | \alpha, \beta, \mu, \delta) d\theta_l$$

end

3. Training:

- Employ the Stochastic Variational Inference model (Stephan et al., 2017).

- Approximate the model on data X and labels y (where applicable, e.g., classification).

- The inference model adjusts model parameters and hyperparameters θ_l in light of the priors $p(\theta_l)$.

Algorithm 1: Fully Bayesian dGP

these challenges by adopting a deep learning strategy: they stack multiple GP layers. Each layer in a dGP functions similarly to a hidden layer in a neural network, transforming the output from the preceding layer—or the input data for the first layer—using its kernel function and hyperparameters. The kernel function k_l at each layer l determines the similarity of outputs for different inputs. Hyperparameters θ_l dictate the kernel's behaviour, influencing aspects like the characteristic length scale and the function's overall scale. The output $f_l(x)$ of layer l is calculated using the transformed output $f_{(l-1)}(x)$ from previous layer, the kernel function k_l , and the hyperparameters θ_l specific to that layer.

We introduce an algorithm (see algorithm 1), for a fully Bayesian dGP model, with priors assigned

to its hyperparameters. The algorithm 1 begins by setting the number of layers L in the dGP structure. For each layer l , ranging from 1 to L , we select a distinct kernel function k_l . This kernel function determines the similarity of the layer’s outputs for various input points. Each kernel function comes with a set of hyperparameters, θ_l , which dictate the kernel’s behaviour, influencing aspects like the smoothness of the function learned or the characteristic length scale of the data. The hyperparameters θ_l are then linked to their respective prior distributions $p(\theta_l)$. We use stochastic variational inference (Hoffman et al., 2013) for parameter estimation. Training involves iteratively adjusting the model parameters and hyperparameters, using the training data X and labels y (when relevant). Throughout this process, the inference engine takes into account the defined priors $p(\theta_l)$ for the hyperparameters, ensuring solutions align with the more probable values according to the priors, thereby regularizing the model and mitigating overfitting.

4 Experiments and Results

The experiments aim to: (1) model the uncertainty inherent in the proposed approach for under-represented languages, (2) evaluate the performance of various multilingual models, and (3) assess the effectiveness of the proposed approach in standard text mining tasks, such as text classification, to show that it can surpass the performance of the nearest benchmark model in this domain.

4.1 Experimental Setup

Datasets: We employed two languages known for their intricate structures: Urdu and Pashto. For Urdu, we used two prominent datasets: the Roman Urdu Emotion Detection Dataset (RUED) (Arshad et al., 2019) and the Urdu Offensive Dataset (UOD) (Akhter et al., 2020). The RUED dataset includes 3075 publicly available instances for emotion detection, labelled as *Anger*, *Sad*, *Happy*, and *Neutral*. Conversely, the UOD dataset is designed for binary classification of offensive language, with 2170 publicly available instances. For Pashto, we utilized the Pashto Offensive Language Dataset (POLD) (Haq et al., 2023), which contains 34400 tweets classified as offensive or non-offensive.

Evaluation Metrics: To evaluate uncertainty, we use Root Mean Square Error (RMSE) and Brier Score metrics (Brier, 1950), commonly adopted in the literature. RMSE helps understand the spread

of errors, providing insight into model uncertainty. The Brier Score calculates the mean squared difference between predicted probabilities (the likelihood of an event) and actual outcomes (whether the event occurred). It penalizes models for prediction errors and overconfidence in wrong predictions. A low Brier Score indicates high accuracy and calibration in predictions, while a high Brier Score suggests lower accuracy or calibration, implying inaccurate predictions or overconfidence, reflecting a lack of understanding of uncertainty. The Brier Score ranges from 0 to 1, with 0 denoting a perfect prediction. For our experiments with binary and multi-class labels, we present the Brier Score (BS) as follows:

$$BS = \underbrace{\frac{1}{D} \sum_{i=1}^D (P_i - O_i)^2}_{\text{Binary}} \& \underbrace{\frac{1}{D} \sum_{i=1}^D \sum_{j=1}^M (f_{ij} - o_{ij})^2}_{\text{Multi}} \quad (2)$$

where D represents the overall number of instances, P_i denotes the predicted probability of the positive class for the i^{th} instance, and O_i signifies the actual outcome (either 0 or 1) for the i^{th} instance, M is the number of classes, f_{ij} represents the forecast probability assigned to class j for instance i . It is the model’s predicted probability that instance i belongs to class j and o_{ij} is the indicator variable for whether class j is the true outcome for instance i . It equals 1 if class j is the true outcome for instance i , and 0 otherwise.

Notice that our methodology is not only adept at uncertainty modelling but can also be used in text classification. We assessed the models’ performance in a few-shot setting (experiments spanned zero-shot, five-shot, fifteen-shot scenarios, and an 80-20 data split). We utilized a 5-fold cross-validation approach and reported the mean values in our results. The main evaluation metric we used is the F1 score.

Language Models: The study employed four multilingual pre-trained transformer models: BERT-base-Multilingual-uncased (Devlin et al., 2018), DistilBERT-base-Multilingual-cased (Sanh et al., 2019), XLM-RoBERTa-base (Conneau et al., 2019), and Multilingual-MiniLM (Wang et al., 2020). These models were selected for their popularity, broad availability, and compatibility with our computational resources. They were fine-tuned over 100 epochs using the Adam optimizer, with an epsilon value of 1e-8 and a learning rate of 2e-5,

incorporating early stopping based on validation loss. The main goals of using these pre-trained models were to create feature vectors and find the best model for the selected languages.

Comparative Model: The study most similar to our model is outlined in the research by (Miok et al., 2022). The authors utilized the Monte Carlo method without priors or kernel functions for uncertainty quantification, closely aligning with the theoretical framework of the deep Gaussian Process model. The primary distinction, however, lies in the fact that their approach does not utilize priors for the model hyperparameters.

Kernel Functions and Hyper-Hyperparameters: The squared kernel, which is a polynomial kernel with an exponent of 2, proved to be the most effective. Despite testing Gaussian, linear, Laplacian, and sigmoid kernels, none outperformed the squared kernel. The performance of all other kernels was approximately 5% lower. The squared kernel’s advantage may be due to its simpler structure and the ease of differentiation, especially when compared to the linear kernel. It also showed greater computational efficiency than more intricate kernels such as the Gaussian and Laplacian.

We utilized conjugate priors (Blei et al., 2003) for mathematical simplicity. Ideally, automating the inference of hyper-hyperparameter values by setting priors, achieved through posterior inference on the hyper-hyperparameters, would be preferred. However, this method incurs a significant computational cost. Research in Bayesian methods such as topic models indicates that setting fixed hyper-hyperparameter values can yield results similar to those from posterior inference on the priors (Blei et al., 2003; Griffiths et al., 2003; Jameel, 2014). In our model, we experimented with two different conjugate priors: the Normal-Inverse-Gamma (NIG) and Student-t distributions. The NIG hyperprior consistently delivered reliable outcomes, in line with previous research (Jameel et al., 2019). NIG priors are well-regarded for providing an optimal balance of flexibility and constraint for hyperparameters, particularly when handling positive hyperparameters and integrating prior knowledge.

4.2 Experimental Results

Uncertainty Modelling: The uncertainty estimation of models prediction on classification tasks is illustrated in Table 2. The primary columns, marked by grey shading, show the results for RMSE and Brier scores. While traditional large

language models are unable to compute Brier and RMSE scores, our proposed method makes this possible. Furthermore, we have benchmarked our method against the baseline model introduced by (Miok et al., 2022) that produced the best results on the BERT-M model. Table 2 presents the performance of leading models across various datasets and test scenarios, including zero-shot, five-shot, fifteen-shot, and full-data. We have showcased the superior quantitative outcomes of the comparative model in our table.

Relative to the baseline model, our method excels in uncertainty modelling, showing superior results in both RMSE and Brier scores. Our model surpasses others for several reasons: it incorporates kernel modelling, which allows it to understand complex relationships within feature vectors, leading to a more precise data representation. This enhances the model’s ability to discern underlying patterns, yielding more accurate predictions and uncertainty assessments. Additionally, our approach includes carefully selected hyper-hyperpriors that steer the learning process by affecting the model’s evaluation of evidence. By strategically choosing them, we ensure the model focuses on the most relevant information, providing reliable and robust uncertainty quantification.

Furthermore, our methodology enabled us to conduct thorough error analysis and pinpoint instances where the model shows confusion and uncertainty in its decision-making. This analysis assists in prioritizing which predictions to scrutinize more closely. Figure 1 illustrates the probability distribution for correct and incorrect predictions by the BERT-Multilingual model in a binary classification task. The distribution is divided into four segments. Predictions nearing 1 signify strong confidence, whereas those approaching 0.5 indicate uncertainty. For a finer view, we also present the 0.6–0.8 range. The graph reveals that predictions made with high confidence (0.8–1.0) tend to be correct, while those made with uncertainty (0.5–0.7) are prone to errors. This highlights our model’s capacity to recognize when it is unsure, allowing us to concentrate on the predictions that require more detailed analysis. Furthermore, our research provides valuable insights that improve uncertainty modelling, especially in multi-class scenarios, enabling the identification of underperforming classes. As illustrated in Figure 2, the Brier score draws attention to the classes that lag in different training setups.

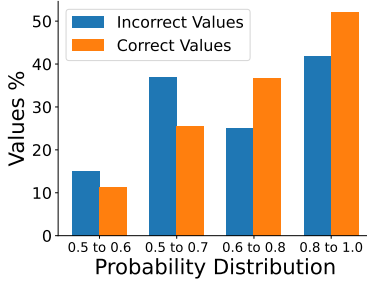


Figure 1: Probability distribution and prediction results for BERT-M for 5 Shot UOD Dataset.

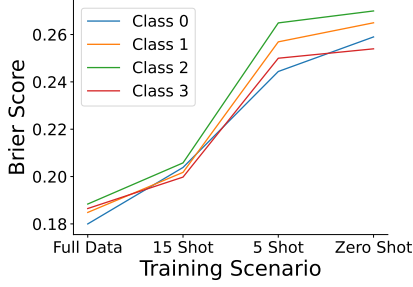


Figure 2: Brier score for each class under different training scenarios for BERT-M in the RUED dataset.

Downstream Application: In addition to uncertainty modeling, our approach is also well-suited for traditional text mining tasks. To demonstrate our model’s superiority over its competitors, we conducted experiments on a downstream application. This study focuses on classification experiments, which aim to sort data points into predefined categories. We evaluated our model using the F1 score metric, as illustrated in Table 2. We tested various multilingual models, addressing both binary and multi-class labels. The table showcases our method’s efficacy in different data availability scenarios: zero-shot, five-shot, fifteen-shot, and full-data. The goal of these experiments was to evaluate the models’ performance on smaller datasets, given the limited public data available for under-represented languages (Khattak et al., 2021).

Deep learning models, such as deep learning-based text classifiers (Minaee et al., 2021), tend to struggle with our datasets due to the underrepresentation of data points that embody complex or subtle variations within a class. These models, lacking sufficient examples to grasp these nuances, may misclassify them or default to the dominant class. Similarly, traditional classifiers like SVMs (Patle and Chouhan, 2013) also face challenges, resulting in a less robust decision boundary that can render the model overly sensitive to certain data points and susceptible to overfitting. Our experi-

Models	UOD (80/20 split)			RUED (80/20 split)			POLD (80/20 split)		
	F1	Brier	RMSE	F1	Brier	RMSE	F1	Brier	RMSE
BERT-M (BL)	0.968	0.029	0.163	0.476	0.189	0.426	0.900	0.079	0.282
BERT-M	0.970	0.025	0.160	0.491	0.185	0.421	0.901	0.078	0.280
XLM-R	0.960	0.038	0.195	0.489	0.189	0.430	0.940	0.073	0.271
DistilBERT	0.947	0.046	0.215	0.467	0.191	0.433	0.861	0.112	0.335
Mini-LM	0.976	0.022	0.148	0.479	0.188	0.424	0.932	0.053	0.230
Models	UOD (15 Shot)			RUED (15 Shot)			POLD (15 Shot)		
	F1	Brier	RMSE	F1	Brier	RMSE	F1	Brier	RMSE
BERT-M (BL)	0.683	0.220	0.451	0.252	0.211	0.448	0.582	0.248	0.497
BERT-M	0.707	0.216	0.449	0.277	0.202	0.443	0.604	0.244	0.494
XLM-R	0.695	0.219	0.452	0.226	0.221	0.461	0.569	0.249	0.498
DistilBERT	0.681	0.239	0.487	0.254	0.201	0.442	0.518	0.260	0.510
Mini-LM	0.671	0.241	0.491	0.221	0.230	0.479	0.539	0.257	0.507
Models	UOD (5 Shot)			RUED (5 Shot)			POLD (5 Shot)		
	F1	Brier	RMSE	F1	Brier	RMSE	F1	Brier	RMSE
BERT-M (BL)	0.611	0.257	0.506	0.247	0.257	0.505	0.375	0.260	0.509
BERT-M	0.620	0.251	0.501	0.251	0.255	0.505	0.397	0.253	0.503
XLM-R	0.524	0.253	0.503	0.188	0.259	0.511	0.360	0.256	0.509
DistilBERT	0.570	0.286	0.547	0.225	0.258	0.509	0.369	0.255	0.507
Mini-LM	0.512	0.291	0.552	0.168	0.267	0.528	0.336	0.261	0.515
Models	UOD (0 Shot)			RUED (0 Shot)			POLD (0 Shot)		
	F1	Brier	RMSE	F1	Brier	RMSE	F1	Brier	RMSE
BERT-M (BL)	0.401	0.262	0.511	0.150	0.265	0.515	0.338	0.269	0.520
BERT-M	0.400	0.258	0.505	0.150	0.263	0.514	0.338	0.267	0.518
XLM-R	0.341	0.261	0.510	0.110	0.268	0.523	0.321	0.272	0.524
DistilBERT	0.353	0.265	0.517	0.137	0.266	0.519	0.333	0.269	0.524
Mini-LM	0.335	0.273	0.569	0.103	0.274	0.533	0.312	0.275	0.531

Table 2: Quantitative Results. BL refers to the results of the closest baseline developed in (Miok et al., 2022).

ments have consistently shown underperformance by these models, which is why they are not included in our experimental results.

Table 2 shows the performance of various multilingual language models on Urdu and Pashto datasets across different scenarios (zero-shot, five-shot, fifteen-shot, and full-data). Across all datasets, our proposed approach performed as well as or better than the baseline model in terms of Accuracy and F1 score. For the UOD dataset, Mini-LM performed best in the full-data scenario, closely followed by BERT-Multilingual, which excelled in the few-shot scenarios compared to other models. Similar trends are observed in the RUED dataset, where BERT-Multilingual performed the best in both full-data and few-shot settings, such as 5-shot and zero-shot. In the POLD dataset, XLM-Roberta demonstrates superior performance in the full-data scenario, closely trailed by Mini-LM. In few-shot scenarios, akin to UOD and RUED datasets, BERT-Multilingual excels in the POLD dataset as well, underscoring its effectiveness.

Comparison with Conventional models: Table 3 provides a detailed comparison of the results between our proposed approach and conventional models. Our model shows comparable or improved performance in terms of F1-score and Accuracy metrics. Notably, unlike conventional models, our model can also compute Brier and RMSE scores, highlighting its additional capabilities in prediction evaluation.

Qualitative Analysis: Based on the experimental evaluation, we have the following key observations.

Our model can effectively identify the instances

Models	UOD (80/20 split)				RUED (80/20 split)				POLD (80/20 split)			
	F1	Acc	Brier	RMSE	F1	Acc	Brier	RMSE	F1	Acc	Brier	RMSE
Conventional Models												
BERT-M	0.958	0.959	-	-	0.510	0.554	-	-	0.901	0.911	-	-
XLN-R	0.960	0.960	-	-	0.473	0.511	-	-	0.940	0.944	-	-
DistilBERT	0.931	0.931	-	-	0.468	0.510	-	-	0.859	0.868	-	-
Mini-LM	0.962	0.964	-	-	0.475	0.530	-	-	0.928	0.941	-	-
Proposed Approach												
BERT-M	0.970	0.970	0.025	0.160	0.491	0.573	0.185	0.421	0.901	0.910	0.078	0.280
XLN-R	0.960	0.959	0.038	0.195	0.489	0.538	0.189	0.430	0.940	0.943	0.073	0.271
DistilBERT	0.947	0.947	0.046	0.215	0.467	0.511	0.191	0.433	0.861	0.870	0.112	0.335
Mini-LM	0.976	0.977	0.022	0.148	0.479	0.530	0.188	0.424	0.932	0.942	0.053	0.230
UOD (15 Shot)				RUED (15 Shot)				POLD (15 Shot)				
Conventional Models												
BERT-M	0.691	0.698	-	-	0.277	0.286	-	-	0.606	0.634	-	-
XLN-R	0.732	0.733	-	-	0.224	0.254	-	-	0.564	0.631	-	-
DistilBERT	0.673	0.676	-	-	0.235	0.325	-	-	0.523	0.630	-	-
Mini-LM	0.665	0.668	-	-	0.201	0.213	-	-	0.536	0.619	-	-
Proposed Approach												
BERT-M	0.707	0.704	0.216	0.449	0.277	0.285	0.202	0.443	0.604	0.650	0.244	0.494
XLN-R	0.695	0.712	0.219	0.452	0.226	0.256	0.221	0.461	0.569	0.630	0.249	0.498
DistilBERT	0.681	0.680	0.239	0.487	0.254	0.331	0.201	0.442	0.518	0.626	0.260	0.510
Mini-LM	0.671	0.670	0.241	0.491	0.221	0.235	0.230	0.479	0.539	0.621	0.257	0.507
UOD (5 Shot)				RUED (5 Shot)				POLD (5 Shot)				
Conventional Models												
BERT-M	0.580	0.583	-	-	0.253	0.261	-	-	0.396	0.433	-	-
XLN-R	0.520	0.527	-	-	0.180	0.254	-	-	0.358	0.410	-	-
DistilBERT	0.573	0.586	-	-	0.211	0.232	-	-	0.370	0.431	-	-
Mini-LM	0.510	0.511	-	-	0.161	0.199	-	-	0.334	0.398	-	-
Proposed Approach												
BERT-M	0.620	0.620	0.251	0.501	0.251	0.273	0.255	0.505	0.397	0.436	0.253	0.503
XLN-R	0.524	0.547	0.253	0.503	0.188	0.258	0.259	0.511	0.360	0.411	0.256	0.509
DistilBERT	0.570	0.589	0.286	0.547	0.225	0.261	0.258	0.509	0.369	0.430	0.255	0.507
Mini-LM	0.512	0.523	0.291	0.552	0.168	0.215	0.267	0.528	0.336	0.399	0.261	0.515
UOD (0 Shot)				RUED (0 Shot)				POLD (0 Shot)				
Conventional Models												
BERT-M	0.340	0.494	-	-	0.150	0.209	-	-	0.335	0.395	-	-
XLN-R	0.328	0.490	-	-	0.111	0.182	-	-	0.323	0.374	-	-
DistilBERT	0.330	0.489	-	-	0.135	0.195	-	-	0.331	0.392	-	-
Mini-LM	0.331	0.486	-	-	0.103	0.164	-	-	0.312	0.360	-	-
Proposed Approach												
BERT-M	0.400	0.517	0.258	0.505	0.150	0.210	0.263	0.514	0.338	0.397	0.267	0.518
XLN-R	0.341	0.502	0.261	0.510	0.110	0.181	0.268	0.523	0.321	0.373	0.272	0.524
DistilBERT	0.353	0.510	0.265	0.517	0.137	0.196	0.266	0.519	0.333	0.394	0.269	0.524
Mini-LM	0.335	0.498	0.273	0.569	0.103	0.165	0.274	0.533	0.312	0.359	0.275	0.531

Table 3: Comparative Results of Proposed Approach with Conventional Models

where it is uncertain, a capability not present in traditional multilingual models. Our method of incorporating priors and kernels increases the model’s confidence in borderline cases, outperforming baseline models. For instance, in predicting the sentiment of the RUED dataset text “Lyari Singoline Me Firing Se 1 Zakhmi [1 injured in firing in Lyari Sangolne]”, the baseline model correctly labelled it as “sad” with Δ is 0.054 ($\Delta = PP - RP$; where PP is the predicted probability and RP is the runner-up probability), which indicates the confusion between the top two probabilities. In contrast, our model not only correctly labelled the text but also

resulted in a higher Δ score of 0.321, demonstrating greater confidence in its prediction.

Since the proposed approach allows us to accurately identify instances of uncertainty, we conducted an additional set of experiments as a case study to strengthen the model’s confidence in areas where uncertainty was detected. We used text from the RUED dataset and leveraged ChatGPT to rephrase ambiguous instances. Our goal was to enhance the sentiment of the text while maintaining its original meaning. The resulting text, referred to as “Advanced Emotion Text”, contains additional emotionally charged words generated by prompt-

Text Type	Passage [Translation]	GL	PL	PP	RP	Δ
Original	khawateen par hone wala tashdud khawateen ki taraqi mein rukawat ka bais banta hai [violence against women is hindrance to women's development]	Sad	Neutral	0.502	0.491	0.011
Advanced Emotion	khawateen par hone wala tashaddud or ziaditi unki taraqqi or khushali mein rukawat ka sabab or waja banti hai [violence and abuse on women is the cause of hindrance in their development and happiness]	Sad	Sad	0.622	0.365	0.257
Extended Emotion	khawateen par hone wala tashaddud na sirf unki taraqqi or khushali mein rukawat ka sabab banta hai, balkih unki honsle ko bhi shikast deta hai, jo samajh ki behteri ke liye bohot zaroori hai [violence against women not only hinders their morals or happiness, but also defeats their courage, which is very necessary for the improvement of society]	Sad	Sad	0.703	0.284	0.419

Table 4: ChatGPT application on a RUED passage with its ground-truth label (GL), predicted labels (PL), predicted probabilities (PP), runner-up probabilities (RP) and Δ (PP-RP) by the BERT-multilingual model.

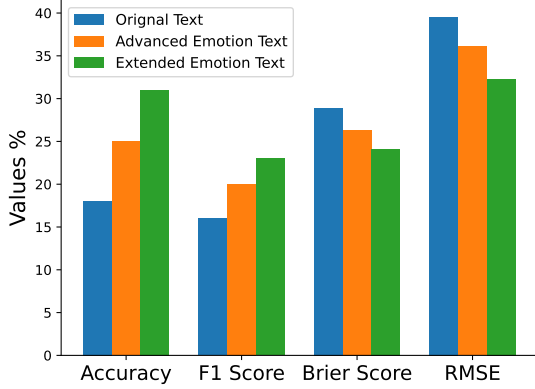


Figure 3: Accuracy, F1, Brier score and RMSE for techniques on less certain predictions.

ing ChatGPT with: “*Enhance the lexicon to elevate the emotional tone of the passages.*” We conducted another experiment where we instructed ChatGPT to rephrase emotive text and lengthen the passages. This resulted in what we call “Extended Emotion Text”. The prompt used for this was: “*Fully enrich the lexicon to intensify the emotional essence of the text and lengthen the sentences.*”

Table 4 shows an example that demonstrates the effectiveness. Initially, the model assigned a low confidence score (small δ value) to an ambiguous passage. However, by applying our proposed techniques, the text was enriched with emotive terms. This allowed the model to accurately predict the category and significantly increase its confidence score (δ value). The key to our approach lies in adding supplementary emotional vocabulary. This leverages the pre-training of our models on emotion datasets, ultimately boosting their confidence and improving their predictive accuracy.

Figure 3 visually illustrates the accuracy, F1 score, RMSE, and Brier score of the mentioned methods when applied to uncertain predictions. The experiments were carried out utilizing the BERT-multilingual model. Significantly, our sug-

gested optimal solution led to a 7% rise in accuracy when employing advanced emotion text and a 13% enhancement with the extended emotion text approach. Moreover, the reduction in Brier score and RMSE indicates improved prediction confidence of the model compared to its previous performance. The effectiveness of this approach lies in its ability to incorporate additional emotional terms, coupled with the pre-training of our models on emotional datasets, which enhances confidence and prediction accuracy. One aspect to bear in mind during experiments is ChatGPT’s inclination to generate hallucinations (Siontis et al., 2024).

5 Conclusions

We have developed a Bayesian dGP model. Our model exerts control over its behaviour including incorporating the kernel functions. The hyper-hyperparameters serve as a meta-level control, shaping the distribution of the standard priors on the hyperparameters. Languages that are less commonly represented possess distinct features or dialectical variations. By employing hyper-hyperparameters, we can steer the model towards learning priors that better suit unseen data, thereby improving its performance on novel language variations. Our experimental results indicate that our model outperforms others in a quantitative comparison.

6 Limitations

The primary challenge for NLP concerning under-represented languages is the lack of data. NLP models require large text datasets for training, but languages with smaller speaker populations often do not have ample written material, online content, or digital resources available. This shortage of data impedes the creation of robust NLP models for these languages. Additionally, these languages often have distinctive linguistic characteristics that

present difficulties for current NLP models. These include intricate morphology, unconventional writing systems, and dialectal differences. Nevertheless, it is vital to develop specialized techniques for these languages.

Our current model achieves satisfactory performance metrics, but we are aware of its limitations and are actively working to improve them. The theoretical advantages of incorporating asymmetric priors are compelling. While these priors could improve model outputs, they also introduce significant computational complexity. However, this complexity can be mitigated by employing techniques such as variational inference, which simplifies the complex posterior distribution into a more tractable form. Methods like the Laplace approximation (Bergamin et al., 2024) or Stochastic Variational Inference (Xuan et al., 2024) can substantially reduce computational costs while maintaining reasonable accuracy. Additionally, we might reduce the kernel size, for instance, by using sparse deep Gaussian processes (Ding et al., 2024) that leverage the natural sparsity of certain kernels, with the assumption that only a limited number of data points have mutual influence. This strategy enables the construction of sparse kernel matrices that are more efficient in terms of storage and computation. Another option is to apply the Nyström Method (Xia, 2024) to approximate the kernel matrix with a low-rank factorization, which can greatly decrease its size and the computational burden for inference.

The selection of priors in Bayesian inference is a subtle process (Martin et al., 2024). In our study, we focus on computational efficiency and mathematical sophistication by employing conjugate priors. Asymmetric priors (Liu and Zhu, 2024) enable the encoding of specific beliefs or knowledge about the process under investigation. This is especially useful when prior knowledge about the expected direction or shape of the variable relationships exists. Incorporating this knowledge via an asymmetric prior can steer the model towards a more accurate and realistic solution. Analogous to the way regularization techniques mitigate overfitting in deep neural networks, asymmetric priors can serve a similar purpose in deep Gaussian Processes. They introduce a preference for simpler models or promote smoothness in predictions, which can prevent the models from becoming excessively complex and overfitting to the data noise.

Unlike conjugate priors (Huang and Huang, 2024), which provide a convenient closed-form

solution for posterior inference, asymmetric priors necessitate more complex numerical methods to approximate the posterior distribution. This complexity can challenge the selection of an appropriate prior that balances the desired influence on the model with computational feasibility. Introducing asymmetric priors can also complicate interpretability, making it more difficult to understand how the prior shapes the posterior distribution and influences the model’s predictions. The increased complexity of inference with asymmetric priors makes it harder to verify and validate the model. Techniques such as cross-validation may require adjustments to consider the prior’s influence on the results.

Moreover, Bayesian methodologies are inherently prone to data quality issues such as biases and outliers (Vieider, 2024). Thus, thorough data cleaning and preprocessing are essential to ensure reliable posterior estimates. Although our specific case has not shown signs of overfitting or underfitting, these issues are potential concerns, especially for Bayesian models with complex structures and limited data. We are exploring regularization techniques to improve the model’s robustness. By recognizing these limitations and actively seeking solutions, we aim for consistent performance enhancements.

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